Data Flow Diagram (DFD): Architecture & Pipeline Telemetry

Project: NeuroClass — Multimodal Attendance & Engagement Agent

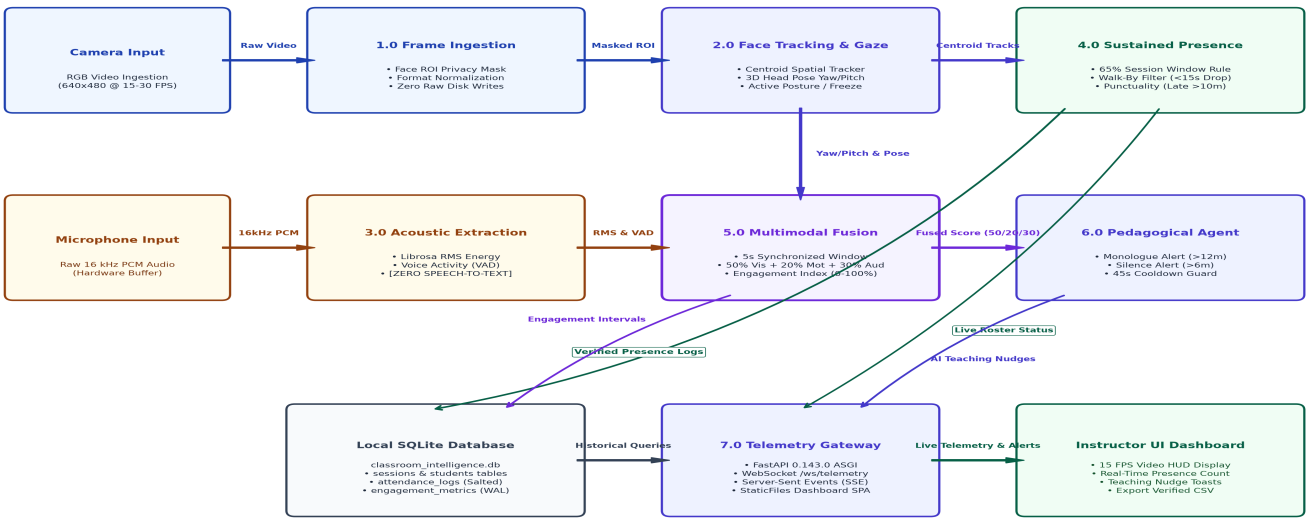
Module: `2.Requirement Analysis`

Phase: Data Flow & Systems Architecture Modeling

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1. Context Diagram (Level 0 DFD)

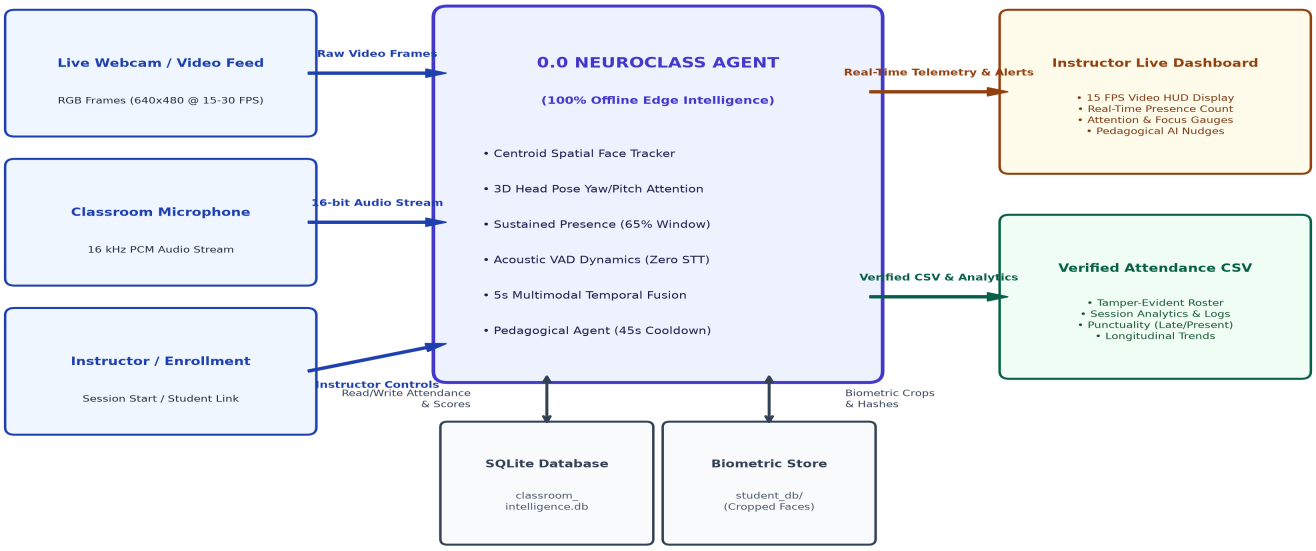
The Level-0 Context Diagram illustrates the boundaries of the NeuroClass system, depicting raw sensory inputs, external actors, and the core data outputs generated by the edge pipeline.



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2. Level 1 Data Flow Diagram (Decomposed Architecture)

The Level-1 Data Flow Diagram breaks down the central system into its seven core interconnected processes:



3. Process Decomposition Specifications

The NeuroClass edge architecture decomposes end-to-end multimodal perception into seven modular, decoupled processing stages. Each process operates asynchronously under strict latency bounds to preserve real-time inference at 15–30 FPS on standard edge hardware without cloud dependencies.

Process 1.0: Frame Ingestion & Privacy Masking

- **Input Data:** Raw RGB/BGR frames from webcam device index or pre-recorded MP4 file (***640 \times 480*** resolution at ***15-30*** FPS).
- **Transformation:**
- Evaluates `privacy_blur_enabled` setting.
- Applies Gaussian/box blurring (***\text{kernel size} = 31***) specifically over detected face bounding boxes.
- Zero raw unmasked frames are saved to disk.
- **Output Data:** Obfuscated frame for display + clean normalized NumPy matrix passed to internal memory trackers.

Process 2.0: Face Tracking & Gaze Estimation

- **Input Data:** In-memory frame matrices.
- **Transformation:**
- Detects faces via Haar cascades / MediaPipe mesh.
- Computes centroid trajectories and assigns stable tracking IDs (`TRK-001`, `TRK-002`).
- Estimates head pose:

$$\text{Yaw} = \frac{\Delta x_{\text{nose-eye}}}{\text{face_width}} \times 90^\circ, \quad \text{Pitch} = \frac{\Delta y_{\text{nose-eye}}}{\text{face_height}} \times 90^\circ$$

- Classifies gaze state: **Attentive** (***|\text{Yaw}| \le 25^\circ \land |\text{Pitch}| \le 20^\circ***) vs. **Distracted**.
- **Output Data:** Coordinate tuples, pose angles, eye openness flags, and tracking tracks.

Process 3.0: Acoustic Feature Extraction

- **Input Data:** 16 kHz raw PCM audio buffer chunks.
- **Transformation:**
- Standardizes waveform via Librosa and calculates Root Mean Square (RMS) energy.
- Computes spectral centroid, spectral rolloff, zero-crossing rate (ZCR), and decibel SPL.
- Runs Voice Activity Detection (VAD) with adaptive noise-floor thresholding (***0.035***).
- Monitors instructor monologue timers (alert triggered at ***>12*** continuous minutes).
- **Zero Speech-to-Text (STT) or transcriptions occur at any step.**
- **Output Data:** `SegmentAcousticFeatures` dictionary containing RMS, VAD boolean, rhythm score, and monologue status.

Process 4.0: Sustained Presence & Attendance Engine

- **Input Data:** Active tracking tracks from Process 2.0 and facial feature embeddings.
- **Transformation:**
- Compares embeddings with pre-enrolled profiles in `student_db/`.
- Enforces **Walk-By Filter:** If active tracking duration ***<15*** seconds, discard as transient passerby.
- Enforces **Sustained Presence Window:**

$$\text{Ratio} = \frac{\text{Observed Active Seconds}}{\text{Total Elapsed Session Seconds}} \ge 0.65$$

- Classifies status: **Present** (***\ge 65\%***), **Late** (first detected after ***>10*** minutes), or **Absent**.
- **Output Data:** Verified attendance events written to SQLite `attendance_logs` table.

Process 5.0: Multimodal Temporal Fusion

- **Input Data:** Visual metrics from Process 2.0 + Acoustic features from Process 3.0.
- **Transformation:**
- Synthesizes metrics across a synchronized 5-second temporal window.
- Applies weighted fusion formula:

$$\text{Score} = (0.50 \times \text{Visual Attention}) + (0.20 \times \text{Motion Balance}) + (0.30 \times \text{Acoustic Interaction})$$

- **Output Data:** Unified Multimodal Representation (UMR) and class-wide Engagement Index (***0-100\%***).

Process 6.0: Pedagogical Reasoning Agent

- **Input Data:** Composite engagement history, monologue alerts, and silence warnings.
- **Transformation:**
- Rule-based fast heuristics + LiteLLM/Groq fallback for pedagogical insights.
- Implements a 45-second suggestion cooldown to prevent teacher alert fatigue.
- Suggests interactive interventions (e.g., "Ask a check-in question", "Take a 2-minute peer discussion pause").
- **Output Data:** Actionable pedagogical suggestion items.

Process 7.0: Telemetry Gateway & Dashboard

- **Input Data:** Live state from all engines + SQLite records.
- **Transformation:**
- Encodes telemetry into JSON payloads transmitted via FastAPI WebSockets (`/ws/telemetry`) and Server-Sent Events (SSE).
- Serves static SPA frontend (`frontend/index.html`) and companion Streamlit interface.
- **Output Data:** Real-time web HUD, interactive charts, and downloadable CSV audit sheets.